

SEM Dendrite Segmentation: Comparative Technical Report

Images evaluated: 10
Failures (Dice<threshold): 10

Average metrics:
Classic Dice=0.925 IoU=0.860 Precision=0.877 Recall=0.979
YOLO Dice=0.000 IoU=0.000 Precision=1.000 Recall=0.000

This PDF was generated by fallback mode (matplotlib) because pdflatex was not available. The matching LaTeX source is summary_report.tex.

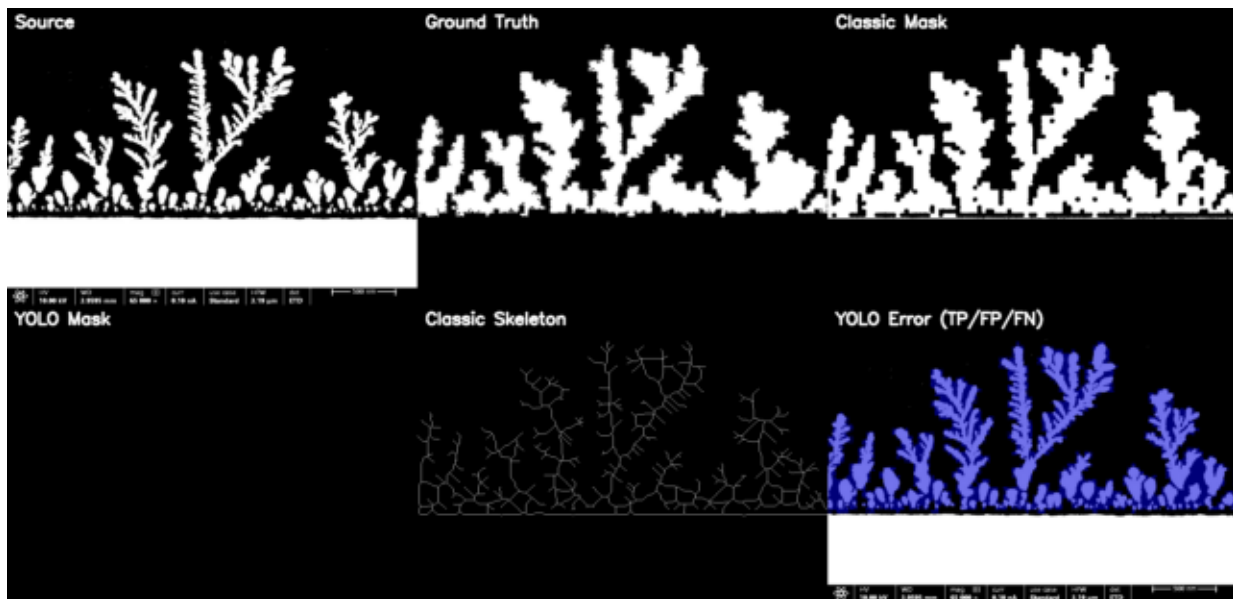
Per-image metrics

Image	C Dice	C IoU	C P	C R	Y Dice	Y IoU	Y P	Y R
easy_1	0.942	0.891	0.903	0.985	0.000	0.000	1.000	0.000
easy_10	0.925	0.860	0.886	0.967	0.000	0.000	1.000	0.000
easy_2	0.930	0.869	0.905	0.956	0.000	0.000	1.000	0.000
easy_3	0.948	0.901	0.922	0.975	0.000	0.000	1.000	0.000
easy_4	0.899	0.817	0.822	0.992	0.000	0.000	1.000	0.000
easy_5	0.917	0.846	0.853	0.991	0.000	0.000	1.000	0.000
easy_6	0.917	0.847	0.855	0.989	0.000	0.000	1.000	0.000
easy_7	0.926	0.862	0.882	0.974	0.000	0.000	1.000	0.000
easy_8	0.909	0.833	0.840	0.990	0.000	0.000	1.000	0.000
easy_9	0.934	0.876	0.897	0.975	0.000	0.000	1.000	0.000

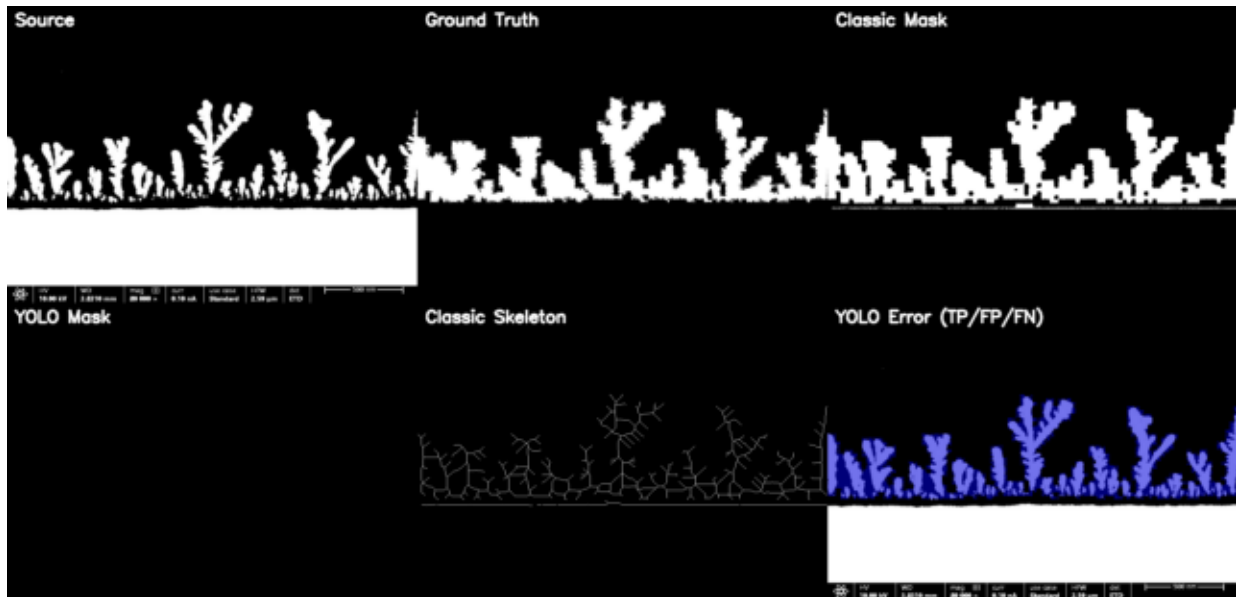
Failure analysis

Image	Method	Dice	Precision	Recall	Cause
easy_1	YOLO	0.000	1.000	Under-segmentation - thin branches missed (Classification)	
easy_10	YOLO	0.000	1.000	Under-segmentation - thin branches missed (Classification)	
easy_2	YOLO	0.000	1.000	Under-segmentation - thin branches missed (Classification)	
easy_3	YOLO	0.000	1.000	Under-segmentation - thin branches missed (Classification)	
easy_4	YOLO	0.000	1.000	Under-segmentation - thin branches missed (Classification)	
easy_5	YOLO	0.000	1.000	Under-segmentation - thin branches missed (Classification)	
easy_6	YOLO	0.000	1.000	Under-segmentation - thin branches missed (Classification)	
easy_7	YOLO	0.000	1.000	Under-segmentation - thin branches missed (Classification)	
easy_8	YOLO	0.000	1.000	Under-segmentation - thin branches missed (Classification)	
easy_9	YOLO	0.000	1.000	Under-segmentation - thin branches missed (Classification)	

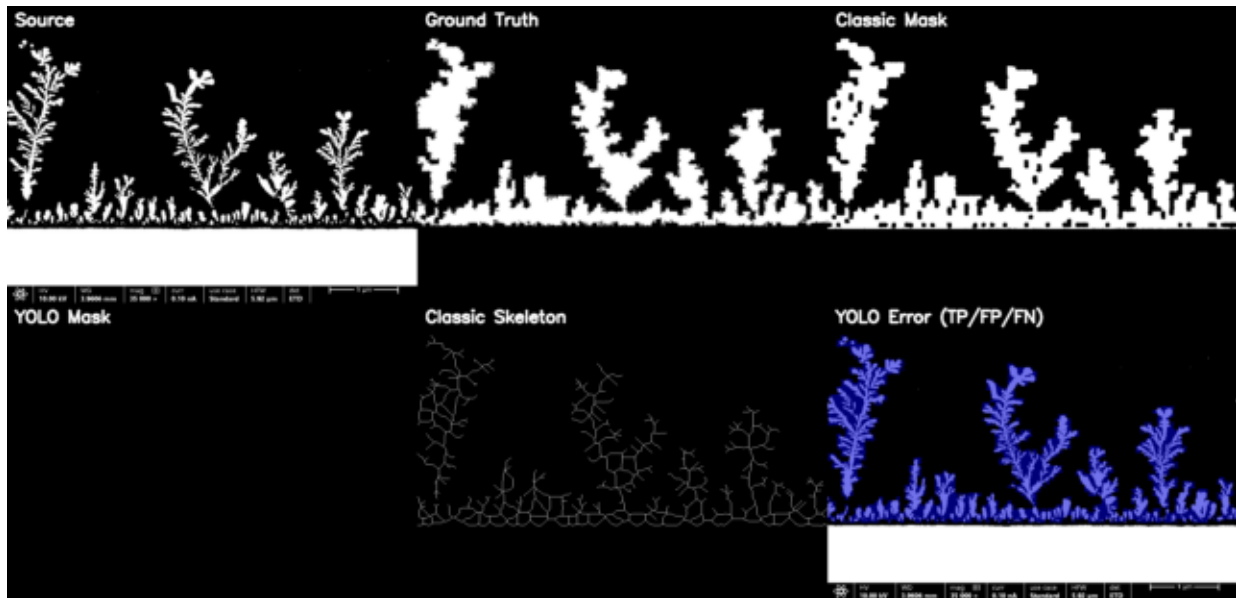
Success case: easy_3 (combined Dice=0.474)



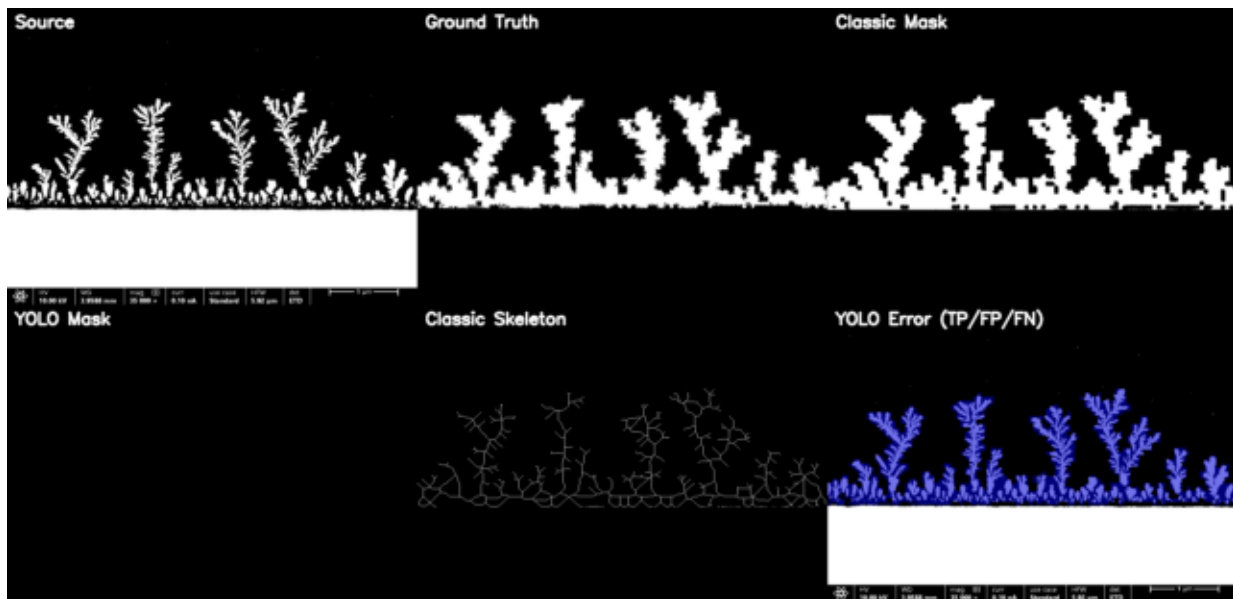
Success case: easy_9 (combined Dice=0.467)



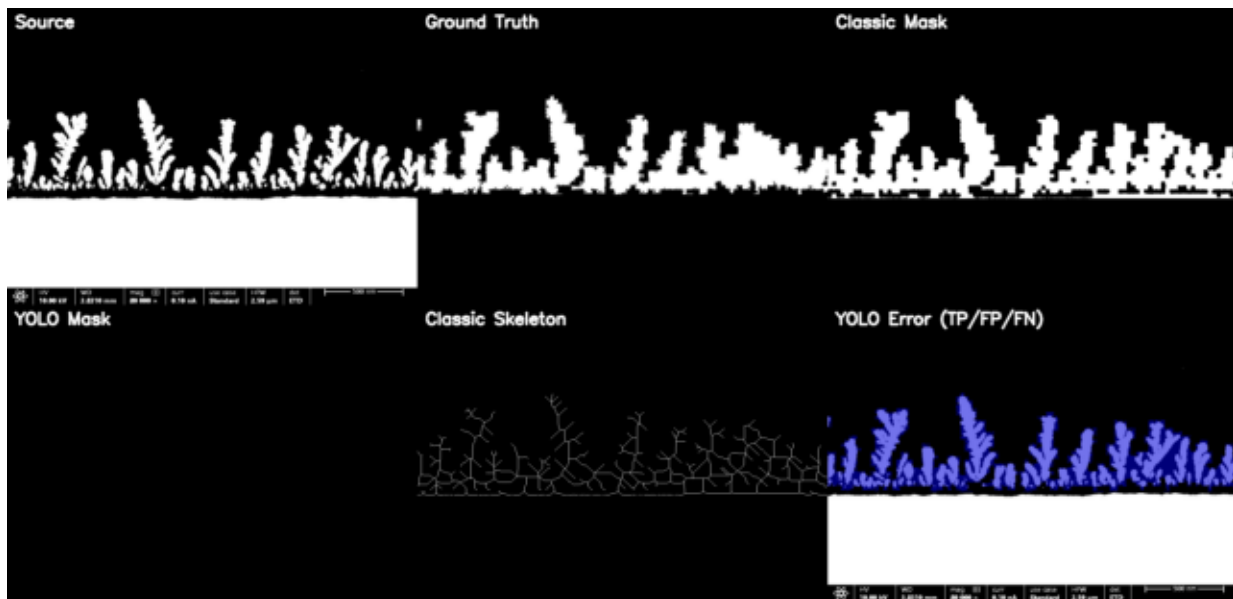
Success case: easy_7 (combined Dice=0.463)



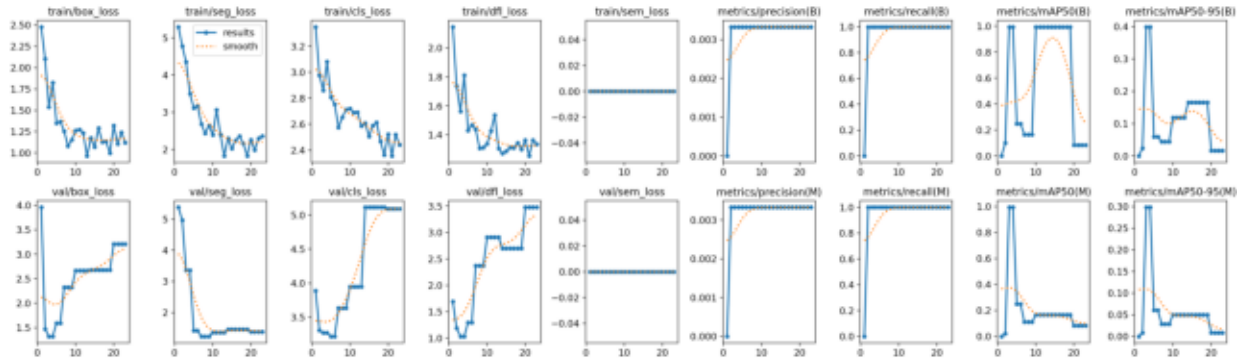
Failure case: easy_1 (combined Dice=0.471)



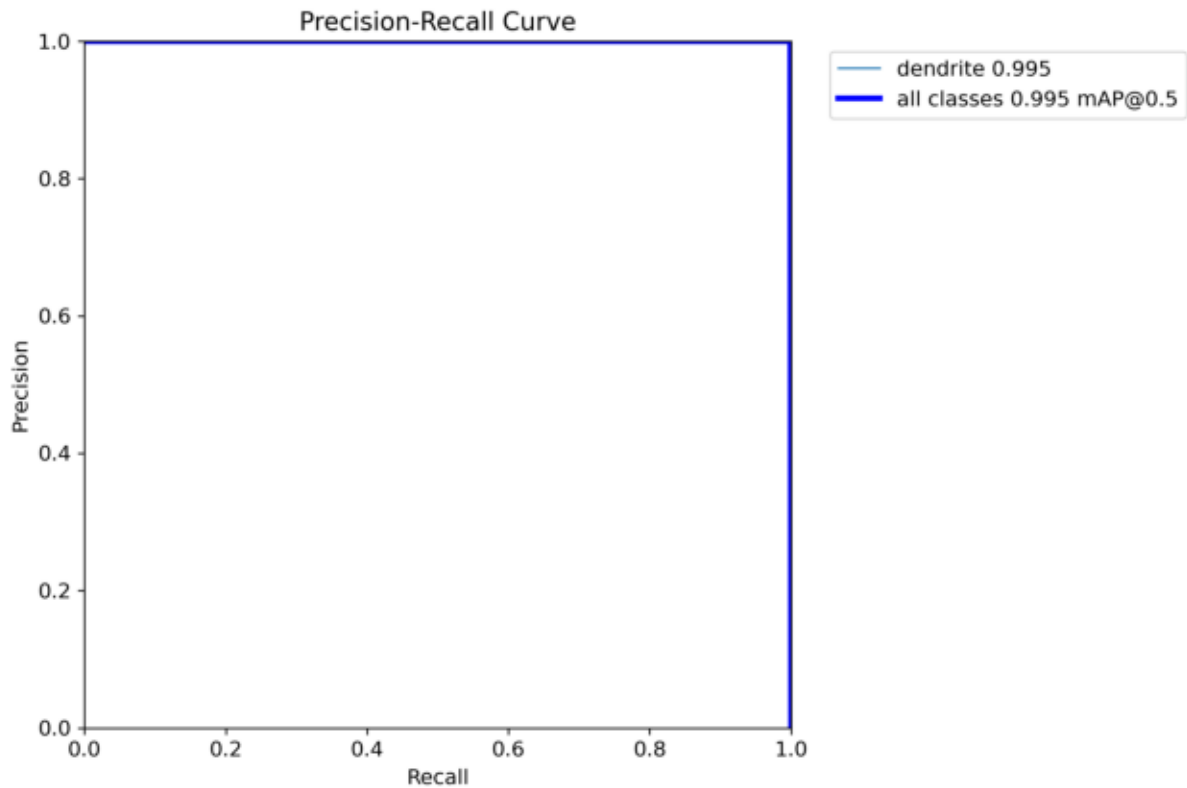
Failure case: easy_10 (combined Dice=0.463)



YOLO training metrics over epochs



Mask precision-recall curve



Normalized confusion matrix

